

Brad Berryman
Site Vice President

Susquehanna Nuclear, LLC
769 Salem Boulevard
Berwick, PA 18603
Tel. 570.542.2904 Fax 570.542.1504
Brad.Berryman@TalenEnergy.com



APR 11 2018

U. S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

10 CFR 50.73

**SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-388(387)/2018-001-00
UNIT 1 LICENSE NO. NPF-14
UNIT 2 LICENSE NO. NPF-22
PLA-7700**

**Docket No. 50-387
50-388**

Attached is Licensee Event Report (LER) 50-388(387)/2018-001-00. This LER is reporting an event which caused a loss of secondary containment differential pressure that was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(v)(C) as a condition that could have prevented fulfillment of a safety function.

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new regulatory commitments.

Should you have any questions regarding this submittal, please contact Mr. Jason Jennings, Manager – Nuclear Regulatory Affairs at (570) 542-3155.

A handwritten signature in blue ink, appearing to be "B. Berryman", with a long horizontal stroke extending to the right.

B. Berryman

Attachment: LER 50-388(387)/2018-001-00

Copy: NRC Region I
Ms. T. E. Hood, NRC Project Manager
Ms. L. H. Micewski, NRC Sr. Resident Inspector
Mr. M. Shields, PA DEP/BRP

Electronic Copy:

B. Berryman
M. M. Thorpe-Kavanaugh (NSRB)
J. R. Jennings
D. J. LaMarca
C. E. Manges, Jr.
T. L. Martin
M. J. Murphy
L. G. Oberrender
G. S. Lubinsky (DBD)

DCS SSES



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOF-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name

Susquehanna Steam Electric Station Unit 2

2. Docket Number

05000388

3. Page

1 OF 3

4. Title Loss of Secondary Containment Differential Pressure During Entry into Unit 2 Zone 3 Exhaust Plenum

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
02	11	2018	2018	001	00	04	11	2018	Susquehanna Steam Electric Station Unit 1	05000387
									Facility Name	Docket Number
										05000

9. Operating Mode

1

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☒ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(iii)
☐ 50.73(a)(2)(i)(C)		☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER

Licensee Contact

C. E. Manges, Jr., Senior Engineer - Nuclear Regulatory Affairs

Telephone Number (Include Area Code)

(570) 542-3089

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To ICES	Cause	System	Component	Manufacturer	Reportable To ICES

14. Supplemental Report Expected

☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No

15. Expected Submission Date

Month Day Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On February 11, 2018 at 22:03, the Susquehanna control room received indication of a loss of Secondary Containment Zone 2 differential pressure and noted differential pressure less than 0.25 inches water gauge (WG) for several seconds. As a result, Technical Specification 3.6.4.1 Surveillance Requirement 3.6.4.1.1 was not met due to loss of Secondary Containment differential pressure and Technical Specification 3.6.4.1 was entered. The alarm came in during an entry into the Unit 2 Zone 3 exhaust plenum for plenum checks. This event is similar to the event that occurred on November 30, 2017 (LER 50-387(388)/2017-007-00).

The condition is being reported in accordance with 10 CFR 50.73(a)(2)(v)(C) as an event or condition that could have prevented fulfillment of a safety function. The condition was previously reported on February 11, 2018, at 23:36 in accordance with 10 CFR 50.72(b)(3)(v)(C) in EN #53202.

This event was unable to be replicated under various scenarios and no obvious equipment issues were identified. A sign was installed on the Unit 2 Zone 3 exhaust plenum door that states: "Entry into the plenum may impact secondary containment, notify control room prior to entry to evaluate LCO requirements."

There were no actual consequences to the health and safety of the public as a result of this event.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Susquehanna Steam Electric Station Unit 2	05000-388	YEAR	SEQUENTIAL NUMBER	REV NO.
		2018	001	00

NARRATIVE**CONDITIONS PRIOR TO EVENT**

Unit 1 – Mode 1, approximately 93 percent Rated Thermal Power

Unit 2 – Mode 1, approximately 100 percent Rated Thermal Power

There were no structures, systems, or components that were inoperable at the start of the event that contributed to the event.

EVENT DESCRIPTION

On February 11, 2018 at 22:03, the Susquehanna control room received indication of a loss of Secondary Containment [EIS System Code: VA] Zone 2 differential pressure and noted differential pressure less than 0.25 inches water gauge (WG) for several seconds. As a result, Technical Specification 3.6.4.1 Surveillance Requirement 3.6.4.1.1 was not met due to loss of Secondary Containment differential pressure and Technical Specification 3.6.4.1 was entered. The alarm came in during an entry into the Unit 2 Zone 3 exhaust plenum for plenum checks. This event is similar to the event that occurred on November 30, 2017 (LER 50-387(388)/2017-007-00).

As a result of the November 2017 event, the Zone 2 filtered exhaust flow controller [EIS System/Component Code: VA/TC] gain and ratio settings were adjusted to have the filtered exhaust fans respond more quickly during a transient. Traces before settings adjustments and after adjustments were reviewed and indicate the adjustment did result in the Zone 2 filtered exhaust fans responding more quickly, but the adjustments did not prevent recurrence of the February 11, 2018 event.

Additional troubleshooting was initiated on February 28, 2018 and March 6, 2018 but was unsuccessful in identifying the cause. Zone 2 differential pressure remained above 0.25 inches WG during the trouble shooting and no obvious component issues were identified.

The condition is being reported in accordance with 10 CFR 50.73(a)(2)(v)(C) as an event or condition that could have prevented fulfillment of a safety function. The condition was previously reported on February 11, 2018, at 23:36 in accordance with 10 CFR 50.72(b)(3)(v)(C) in EN #53202.

CAUSE OF EVENT

This event was unable to be replicated and no obvious equipment issues were identified during troubleshooting.

ANALYSIS/SAFETY SIGNIFICANCE

Based on engineering analysis of the event, secondary containment could have performed its safety function of isolating as assumed in the accident analysis and of re-establishing 0.25 inches of vacuum WG (drawdown) within the assumed accident analysis time (10 minutes).

This event will not be counted as a safety system functional failure (SSFF) for the NRC performance indicator based on the engineering analysis supporting the system's ability to fulfill the safety function.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOF-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
Susquehanna Steam Electric Station Unit 2		05000-388	YEAR 2018	SEQUENTIAL NUMBER 001	REV NO. 00

CORRECTIVE ACTIONS

Corrective actions include the following:

1. A sign was installed on the Unit 2 Zone 3 exhaust plenum door that states: "Entry into the plenum may impact secondary containment, notify control room prior to entry to evaluate LCO requirements."

PREVIOUS SIMILAR EVENTS

The following is a recent related LER involving loss of secondary containment:

LER 50-387(388)/2017-007-00, "Secondary Containment Declared Inoperable Due to the opening of a plenum," dated January 25, 2018.